

Neonatal diagnosis of biliary atresia: a practical review and update.

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Biliary atresia is challenging to diagnose because many of the clinical and imaging features of this condition overlap with those of other causes of cholestasis in newborns. When jaundice persists beyond 2 weeks of age, the neonate should be evaluated for cholestasis, and biliary atresia - the most common cause of neonatal cholestasis - should be considered. It is critical to diagnose biliary atresia early because failure to treat can result in hepatic fibrosis and death in less than 1 year. In this paper, we review the current diagnostic imaging methods, differential considerations and treatment options for biliary atresia.

Population-based screening strategies for biliary atresia in the newborn: A systematic review and meta-analysis.

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Newborn screening for biliary atresia (BA) may facilitate earlier diagnosis and intervention for improved clinical outcomes. We systematically reviewed the accuracy of population-based screening strategies for BA in the newborn using PRISMA-DTA guidelines. We included cohort or cross-sectional studies. The screening (index) tests included stool color card (SCC) and direct/conjugated bilirubin (DB/CB) and the reference standard was intraoperative cholangiogram. Meta-analysis was performed using random-effects logistic regression models. We included 15 studies (1,816,722 participants) that assessed 5 different population-based screening strategies. QUADAS-2 assessment revealed high risk of bias for patient selection in one study and uncertain risks for reference standard in multiple studies. High certainty evidence suggests that DB/CB assessed after birth had a summary sensitivity of 100% (95% CI 100,100) and specificity of 98.8% (98.8,98.9) (5 studies, 662141 participants). Moderate certainty evidence suggests that SCC screening at a month of age had summary sensitivity of 79.6% (95% CI 70.6, 86.4) and specificity of 99.9% (95% CI 99.9, 99.9) (7 studies, 996262 participants). DB/CB in the first few days of life has the best diagnostic accuracy for population screening for biliary atresia in the newborn. Future research should focus on cost-effectiveness and combinations of screening strategies.

Psychological Journey and Coping Styles of Parents of Infants With Biliary Atresia: A Single-Center Qualitative Study.

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Biliary atresia is a rare and serious neonatal disease that affects the quality of life of both infants and parents. There is currently limited literature on the experiences of parents with infants diagnosed with biliary atresia. To explore the psychological journey and coping styles of parents of infants with biliary atresia in a single center in Shanghai, China. A qualitative study design was used. Face-to-face and semistructured interviews were conducted with 10 parents of infants with biliary atresia. Colaizzi's method of data analysis was utilized, using NVivo 11.0 software. The psychological journey and coping styles of parents could be divided into 4 stages. Different themes were extracted at different stages: before diagnosis, parents experienced complex emotions and actively sought treatment; at the diagnosis stage, negative emotions dominated and parents convinced themselves to accept reality; in the postoperative stage, positive emotions, accepting reality, active response, and the need to learn to take

care of their infant gradually appeared; and at the discharge stage, parents accepted the coexistence of positive and negative emotions and the variety of needs that emerged. The findings of the study may help healthcare professionals identify and focus on the psychological needs of parents of infants with biliary atresia, leading them to implement effective coping strategies to increase the caregiving ability of parents. Future research should explore the effects of supportive interventions for parents of infants with serious chronic illnesses.

Polymeric immunoglobulin receptor promotes Th2 immune response in the liver by increasing cholangiocytes derived IL-33: a diagnostic and therapeutic biomarker of biliary atresia.

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Biliary atresia (BA) is a devastating neonatal cholangiopathy with an unclear pathogenesis, and prompt diagnosis of BA is currently challenging. Proteomic and immunoassay analyses were performed with serum samples from 250 patients to find potential BA biomarkers. The expression features of polymeric immunoglobulin receptor (PIGR) were investigated using human biopsy samples, three different experimental mouse models, and cultured human biliary epithelial cells (BECs). Chemically modified small interfering RNA and adenovirus expression vector were applied for in vivo silencing and overexpressing PIGR in a rotavirus-induced BA mouse model. Luminex-based multiplex cytokine assays and RNA sequencing were used to explore the molecular mechanism of PIGR involvement in the BA pathogenesis. Serum levels of PIGR, poliovirus receptor (PVR), and aldolase B (ALDOB) were increased in BA patients and accurately distinguished BA from infantile hepatitis syndrome (IHS). Combined PIGR and PVR analysis distinguished BA from IHS with an area under the receiver operating characteristic curve of 0.968 and an accuracy of 0.935. PIGR expression was upregulated in the biliary epithelium of BA patients; Th1 cytokines TNF- α and IFN- γ induced PIGR expression in BECs via activating NF- κ B pathway. Silencing PIGR alleviated symptoms, reduced IL-33 expression, and restrained hepatic Th2 inflammation in BA mouse model; while overexpressing PIGR increased liver fibrosis and IL-33 expression, and boosted hepatic Th2 inflammation in BA mouse model. PIGR expression promotes the proliferation and epithelial-mesenchymal transition, and reduced the apoptosis of BECs. PIGR participated in BA pathogenesis by promoting hepatic Th2 inflammation via increasing cholangiocytes derived IL-33; PIGR has the value as a diagnostic and therapeutic biomarker of BA. This study was financially supported by the National Natural Science Foundation of China (82170529), the National Key R&D Program (2021YFC2701003), and the National Natural Science Foundation of China (82272022).

Development and validation of a model for early diagnosis of biliary atresia.

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Early diagnosis of biliary atresia (BA), particularly distinguishing it from other causes of neonatal cholestasis (NC), is challenging. This study aimed to design and validate a predictive model for BA by using the data available at the initial presentation. Infants presenting with NC were retrospectively identified from tertiary referral hospitals and constituted the model design cohort (n = 148); others were enrolled in a prospective observational study and constituted the validation cohort (n = 21). Clinical, laboratory, and abdominal ultrasonographic features associated with BA were assessed. A prediction

model was developed using logistic regression and decision tree (DT) analyses. Three predictors, namely, gamma glutamyl transpeptidase (γ GT) level, triangular cord sign (TC sign), and gallbladder abnormalities, were identified as factors for diagnosing BA in multivariate logistic regression, which was used to develop the DT model. The area under the receiver operating characteristic (ROC) curve (AUC) value for the model was 0.905, which was greater than those for γ GT level, TC sign, or gallbladder abnormalities alone in the prediction of BA. A simple prediction model combining liver function and abdominal ultrasonography findings can provide a moderate and early estimate of the risk of BA in patients with NC.

Amyloid precursor protein as a fibrosis marker in infants with biliary atresia.

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Biliary atresia (BA) is a rare condition of unknown origin in newborns with jaundice. In BA bile ducts are non-functional, causing neonatal cholestasis and following liver fibrosis and failure. This retrospective study included liver biopsies of 14 infants with BA aged [mean $\pm$ SD] 63 ± 23 days. Patients were grouped according to the clinical course (jaundice-free vs recurrent jaundice vs required liver transplantation or liver fibrosis (Ishak fibrosis score)) and followed for 1.61-5.64 years (mean 4.03). Transcriptome profiles were assessed using a panel of 768 fibrosis-specific genes, reanalyzed via qRT-PCR, and confirmed via immunostaining. Plasma from an additional 30 BA infants and 10 age-matched controls were used for amyloid precursor protein (APP) quantification by ELISA. Different clinical outcome groups showed a homogeneous mRNA expression. Altered amyloid-metabolism-related gene expression was found between cases with Ishak fibrosis score greater than 4. Immunostaining confirmed a distinct presence of APP in the livers of all BA subjects. APP plasma levels were higher in BA than in age-matched controls and correlated with the histological fibrosis grade. These results suggest that amyloidosis may contribute to BA and liver fibrosis, indicating that APP could serve as a potential liquid biomarker for these conditions. Biliary atresia patients with higher fibrosis scores according to Ishak have higher hepatic expression of amyloid-related genes while amyloid precursor protein accumulates in the liver and increases in the circulation. After a recent study revealed beta-amyloid deposition as a mechanism potentially involved in biliary atresia, we were able to correlate amyloid-metabolism-related transcript levels as well as amyloid precursor protein tissue and plasma levels with the degree of hepatic fibrosis. These findings suggest that amyloid precursor protein is a fibrosis marker in infants with biliary atresia, reinforcing the role of amyloid metabolism in the pathogenesis of this serious disease.

Porta hepatis lymphnode mimicking biliary atresia: A case report.

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Cholestasis is the impairment of normal bile flow causing accumulation of bile salts, lipids, and bilirubin in blood which presents as Jaundice. Jaundice beyond 2 weeks of age is rare in infancy with worldwide incidence of 1 in 2500 live births. Biliary atresia is the most common extra hepatic cause of cholestasis in late neonatal and infancy period. Cholestasis and hyperbilirubinemia cause irreversible brain and liver damage if not diagnosed and treated early. A 3-week-old neonate presenting with progressive yellowish discoloration of eyes and skin. Explorative laparotomy found anatomically normal liver and biliary tree, but a lymph node obstructing the common bile duct. This case was particularly unique as history of illness and initial investigations were suggestive of biliary atresia. However, the patient had lymph nodes

with no history of any triggers to lymphadenopathy. It is a rare case of obstruction of biliary flow in this age group. Despite biliary atresia being the commonest cause of obstructive jaundice in infancy, it is important to rule out other causes like lymph nodes obstructing the biliary tree.

Biliary atresia.

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Biliary atresia (BA) is a rare disease characterised by a biliary obstruction of unknown origin that presents in the neonatal period. It is the most frequent surgical cause of cholestatic jaundice in this age group. BA occurs in approximately 1/18,000 live births in Western Europe. In the world, the reported incidence varies from 5/100,000 to 32/100,000 live births, and is highest in Asia and the Pacific region. Females are affected slightly more often than males. The common histopathological picture is one of inflammatory damage to the intra- and extrahepatic bile ducts with sclerosis and narrowing or even obliteration of the biliary tree. Untreated, this condition leads to cirrhosis and death within the first years of life. BA is not known to be a hereditary condition. No primary medical treatment is relevant for the management of BA. Once BA suspected, surgical intervention (Kasai portoenterostomy) should be performed as soon as possible as operations performed early in life is more likely to be successful. Liver transplantation may be needed later if the Kasai operation fails to restore the biliary flow or if cirrhotic complications occur. At present, approximately 90% of BA patients survive and the majority have normal quality of life.

[First orthotopic liver transplantation in patient with biliary atresia and situs inversus in Spain].

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Biliary atresia is the most common indication for liver transplantation in the pediatric age group. The Kasai portoenterostomy has become established as the primary treatment for biliary atresia. If portoenterostomy fails, death before 2 years of age is likely without liver transplantation. The most common multiple malformation syndrome associated with biliary atresia is polysplenia syndrome, which forms a constellation of defects of body symmetry, splenic development and vascular anomalies, including situs inversus, polysplenia and others. The situs inversus was formerly considered an absolute contraindication for liver transplantation. Recently however, several case reports have been published suggesting that neither situs inversus nor this particular subset of vascular abnormalities should be considered contraindications to liver transplantation. We present one case of liver transplantation performed in patient with biliary atresia, situs inversus and polysplenia. This is the first report described in Spain for a liver transplant in a child with biliary atresia plus situs inversus.

Newborn Screening for Biliary Atresia.

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Biliary atresia is the most common cause of pediatric end-stage liver disease and the leading indication for pediatric liver transplantation. Affected infants exhibit evidence of biliary obstruction within the first

few weeks after birth. Early diagnosis and successful surgical drainage of bile are associated with greater survival with the child's native liver. Unfortunately, because noncholestatic jaundice is extremely common in early infancy, it is difficult to identify the rare infant with cholestatic jaundice who has biliary atresia. Hence, the need for timely diagnosis of this disease warrants a discussion of the feasibility of screening for biliary atresia to improve outcomes. Herein, newborn screening for biliary atresia in the United States is assessed by using criteria established by the Discretionary Advisory Committee on Heritable Disorders in Newborns and Children. Published analyses indicate that newborn screening for biliary atresia by using serum bilirubin concentrations or stool color cards is potentially life-saving and cost-effective. Further studies are necessary to evaluate the feasibility, effectiveness, and costs of potential screening strategies for early identification of biliary atresia in the United States.
